Pranaykumar Tirpude

Email: pranaykt@udel.edu
Web: https://frozenarchives.github.io

University of Delaware
Department of Earth Sciences
255 Academy St, Newark, DE 19716

Research Interests

Antarctic Ice Sheet dynamics, Pleistocene climate variability, Ice-Rafted Debris (IRD) records, machine learning applications in paleoclimate, isotope geochemistry, and ice sheet and iceberg trajectory modeling.

Education

Ph.D. in Earth Sciences, University of Delaware (Aug 2023 – Present)
Advisor: Dr. Chandranath Basak
Dissertation: Antarctic Ice Sheet Dynamics across the Mid-Pleistocene Transition Through High-Resolution Ice-Rafted Debris Records

M.Sc. in Geology, Banaras Hindu University (Jul 2021)
Thesis: AI/ML-Based Sea Surface Temperature Estimation Using Planktic Foraminiferal Assemblage from IODP Site U1457
Advisor: Prof. Arun Deo Singh

B.Sc. in Geology, Fergusson College Pune (Jun 2019)

Research Experience

Ph.D. Candidate, University of Delaware, Dept. of Earth Sciences (2023 – Present)

- Project: Antarctic Ice Sheet Dynamics across the Mid-Pleistocene Transition through high-resolution IRD records.
- Conducting deep learning-based instance segmentation of IRD from X-ray scans.
- Time series analysis of IRD flux spanning 1.4 million years.
- Developing ice sheet and iceberg-trajectory models to understand Antarctic ice sheet dynamics and IRD delivery to ice-distal locations.
- Antarctic IRD provenance studies using Strontium (Sr), Neodymium (Nd), and Lead (Pb) isotopes.

Summer Visitor, NSF National Center for Atmospheric Research (NCAR), Boulder, CO (Jun 2026 – Jul 2026)

- Project: Linking Southern Ocean Ice-Rafted Debris Flux with Antarctic Ice Sheet Dynamics
- Simulate Antarctic ice sheet calving rates for the modern and past climates using the Community Ice Sheet Model (CISM)
- Generate iceberg size distribution from the simulated CISM calving rates
- Iceberg trajectory modeling including breakup of giant icebergs and meltwater distribution using generated icebergs
- Validating the ice sheet and iceberg trajectory model outputs by radiogenic isotopic analyses of IRD grains

Junior Research Fellow, Indian Institute of Science (2021 – 2023)

- Project: Reconstruction of atmospheric CO_2 using Boron isotope ratios in marine carbonates.
- Performed species-level identification of foraminifera.
- Executed chemical cleaning and processing of samples in a class-100 clean lab.
- Operated ICP-OES and triple quadrupole ICP-MS instruments.

M.Sc. Student, Banaras Hindu University (2019 – 2021)

- Project: Sea surface temperature estimation using machine learning and foraminifera.
- Applied ANN, SVM, AdaBoost, Random Forest, and other ML techniques.
- Coding experience in Python, C++, BASH, and SLURM; proficient in Linux, Windows, and MacOS.
- Experience with PyFERRET, PyGMT, SeaDAS, and Ocean Data View (ODV).

Honors and Awards

- **Geological Society of America Graduate Student Research Grant**
  Research grant awarded to conduct radiogenic isotope analysis on whole rock Antarctic IRD grains (Jul 2026)
- **Climate Hub – Lundgaard Awards,** University of Delaware, Apr 2026
  Awarded to support a two-month research visit to NSF NCAR in Boulder, CO
- **Geological Society of America Travel Grant,** Mar 2026
  Awarded to support travel to attend the GSA Northeastern meeting in Hartford, Connecticut
- **ONGC Scholarship for meritorious students in geosciences**
  Awarded by Oil and Natural Gas Corporation of India (Dec 2019).
- **INSA Summer Research Fellowship**
  Jointly awarded by the Indian National Science Academy, Indian Academy of Science, and the National Academy of Sciences, India to support selected

undergraduate students to conduct summer research in top research institutions in India (Jun 2018).

Seagoing Experience

SEALS 2025 (Sediment Exchange Along the Labrador Sea) (June – July 2025)

- Seagoing research expedition aboard R/V Roger Revelle with 29 scientists from 12 different institutions
- Conducted seawater sampling using a CTD rosette, assisted with multi-corer and gravity coring, porewater sampling, and alkalinity titration.
- UDaily article: https://www.udel.edu/udaily/2025/september/seals-research-expedition-nsf-rare-earth-elements-labrador-sea/

Teaching Experience

Teaching Assistant, University of Delaware (Fall 2023 – Spring 2024)

- Taught lab sections for GEOL107 – Geology of Dynamic Earth, GEOL113 – Earth Science, and GEOL115 – Geological Hazards.
- Assisted students with hands-on lab activities, assignments, and course concepts.

Conferences, Workshops, and Internships

- **Oral Presentation:** Linking Southern Ocean Ice-Rafted Debris Flux to Antarctic Ice Sheet Dynamics, Community Earth System Model (CESM) Workshop, Jul 2026 (NSF NCAR, Boulder, CO)
- **Poster Presentation:** Can an ice-distal IODP site capture signals of West Antarctic Ice Sheet Dynamics across the Mid-Pleistocene Transition? Insights from the South Pacific (Geological Society of America Northeastern Section, Hartford, CN, Mar 2026)
- **Oral Presentation**: Understanding the Antarctic Ice Sheet Dynamics Through High-Resolution IRD Records Using AI/ML Models, American Geophysical Union (Washington, DC, Dec 2024) (Link).
- **Workshop**: Python and R Analysis of Time Series and FAIR Science Publishing, PyRATES/LinkedEarth, USC Information Sciences Institute (Fully funded, Jun 2024) (Link).
- **Summer Research Fellowship**: Micropaleontological investigation of Deccan Intertrappean beds, University of Delhi (Jun 2018).
- **Winter Internship**: Lithostructural mapping and radiometric survey in uranium mineralized zones of eastern India, Atomic Minerals Directorate, Department of Atomic Energy, Govt. of India (Dec 2019).

Outreach Activities

- **Coast Day, University of Delaware** (Oct 2023, 2024): Demonstrated the significance of scientific ocean drilling and paleoclimate studies to ~200 participants.
- **IISc Open Day**, Indian Institute of Science (Feb 2023): Demonstrated the role of microfossils in paleoclimate research to students and the general public.

Memberships

- American Geophysical Union (AGU).
- Geological Society of America (GSA).